

Finding Seismic Discontinuities in Mantle Circulation Models: the MTZ and beyond? *Supplementary Figures*



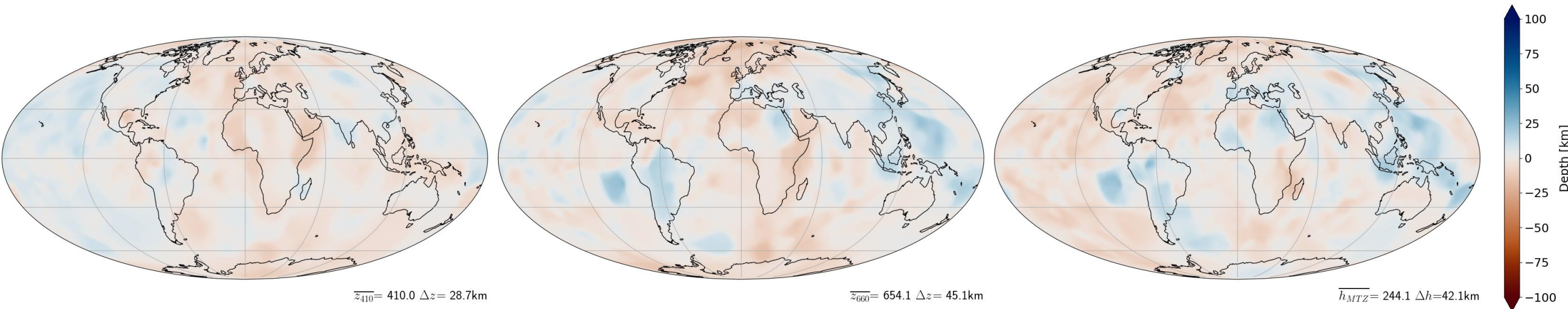
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James Panton^[1] * Presenting Author: morgangt2@cardiff.ac.uk

^[1]School of Earth and Environmental Sciences, Cardiff University

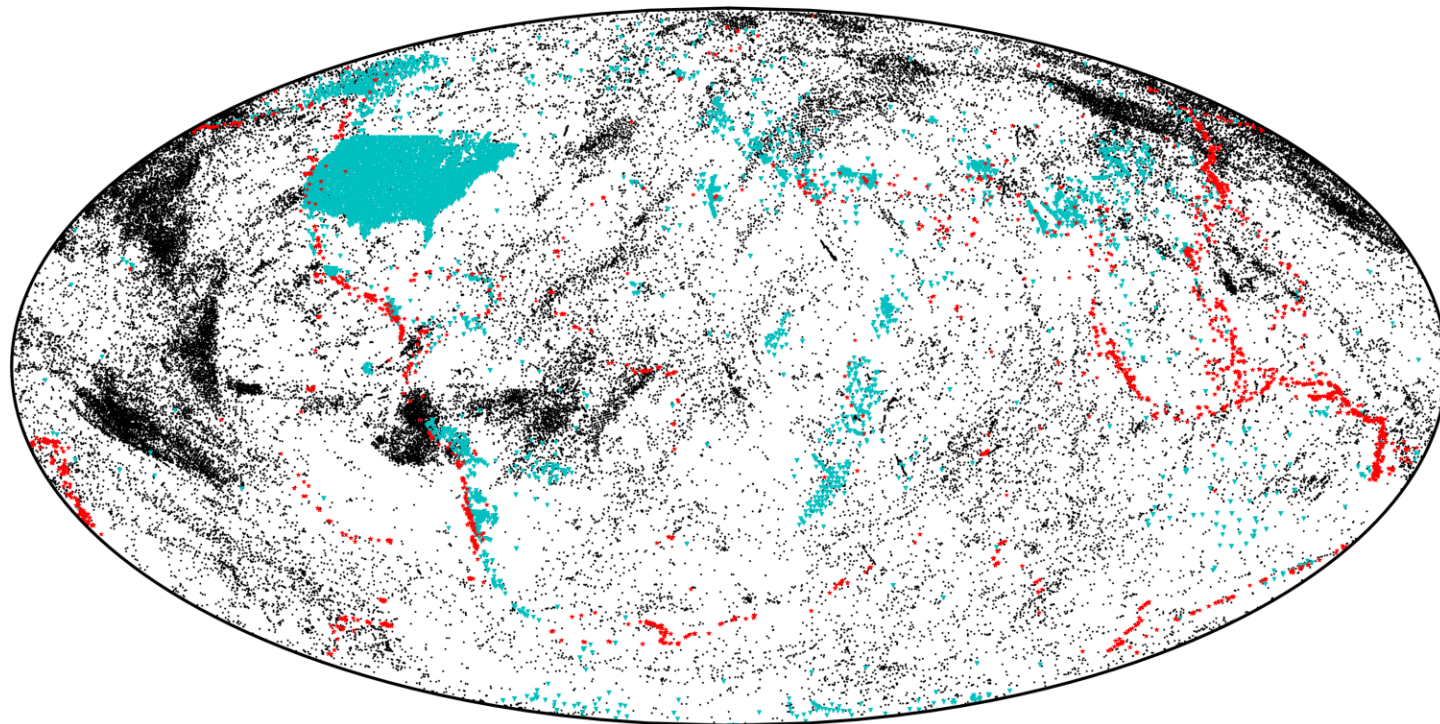
^[2]School of Earth Sciences, University of Bristol

Waszek et al 2021

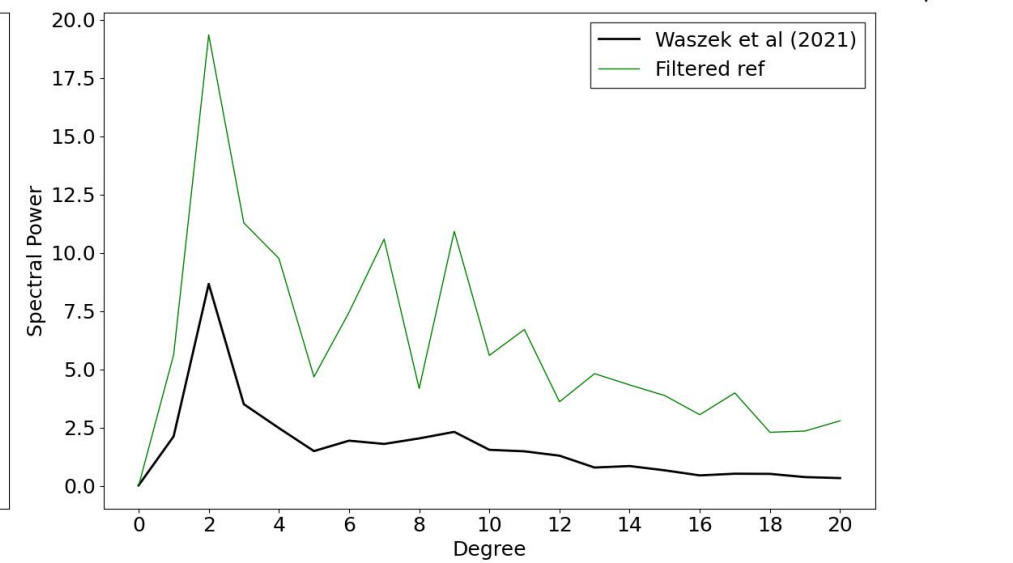
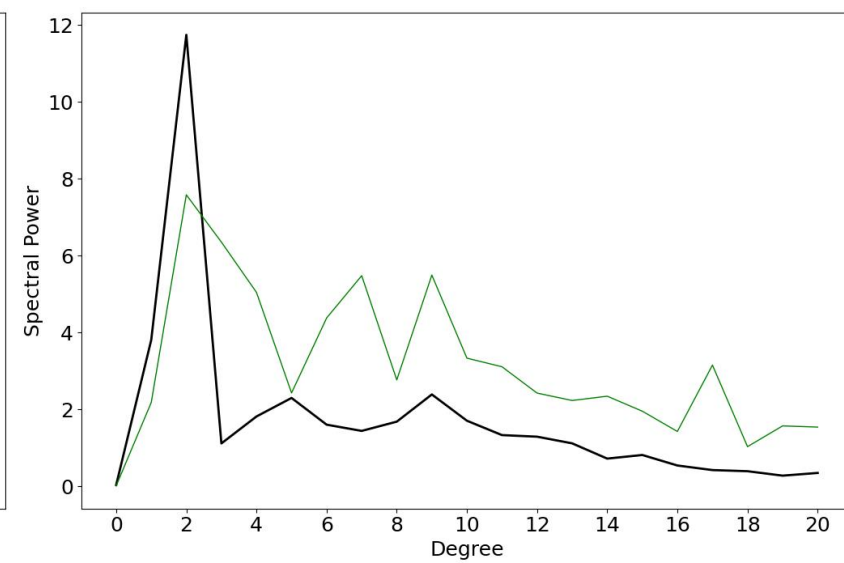
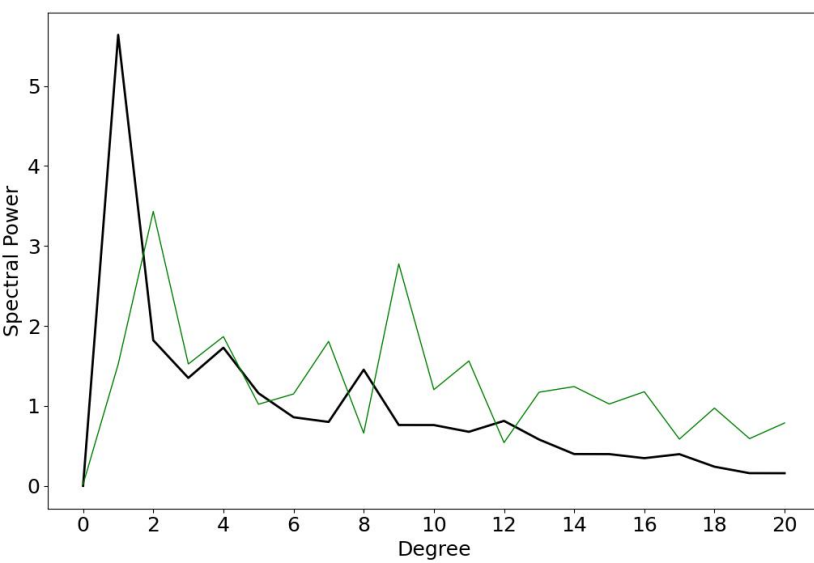
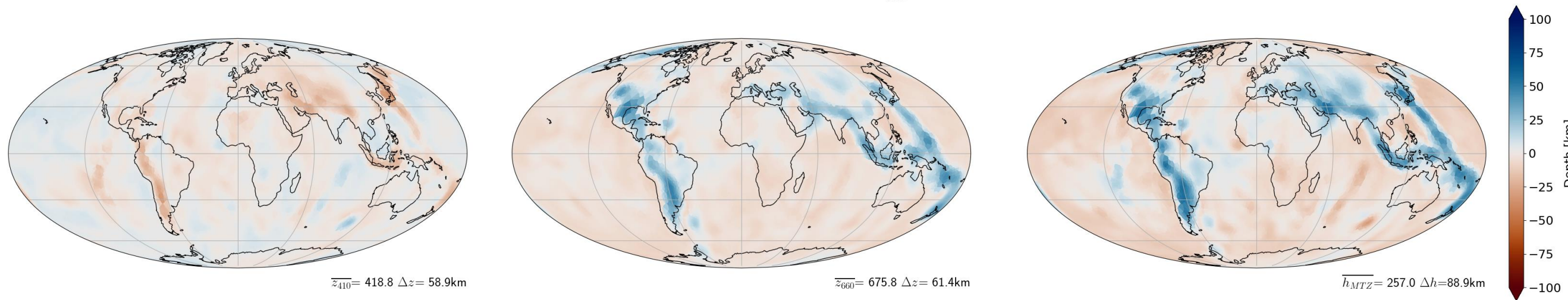
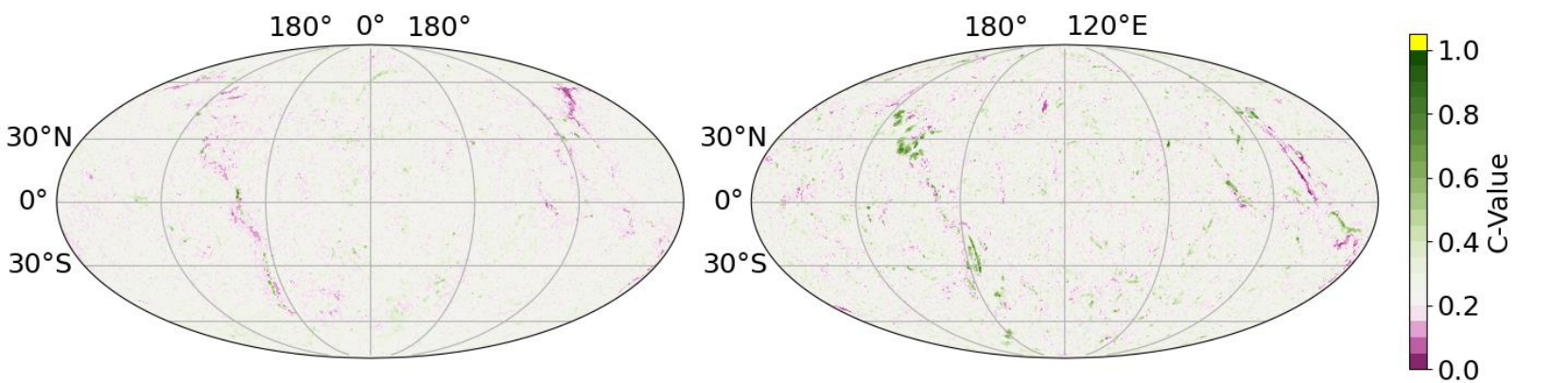
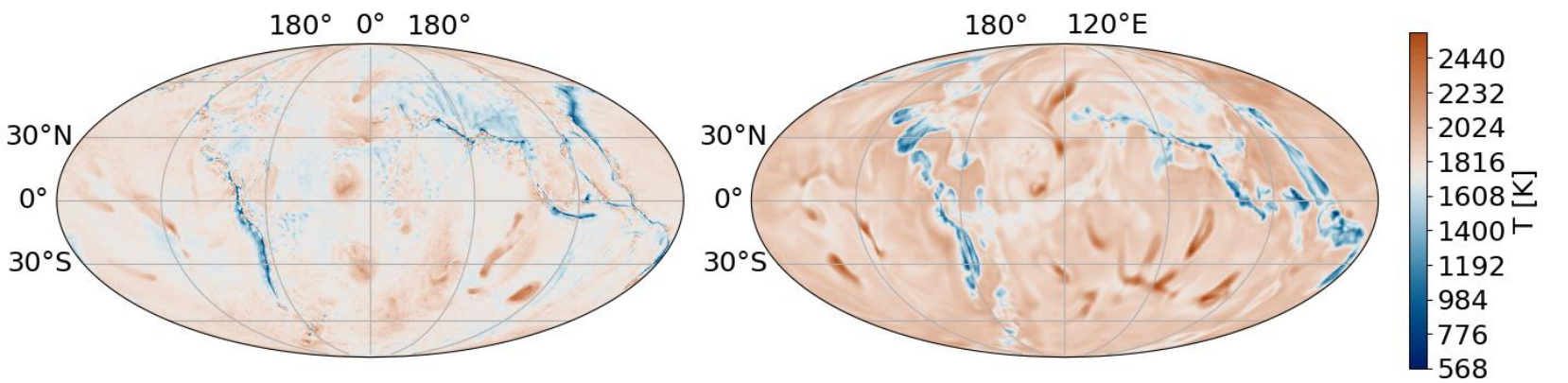
MTZ topographies



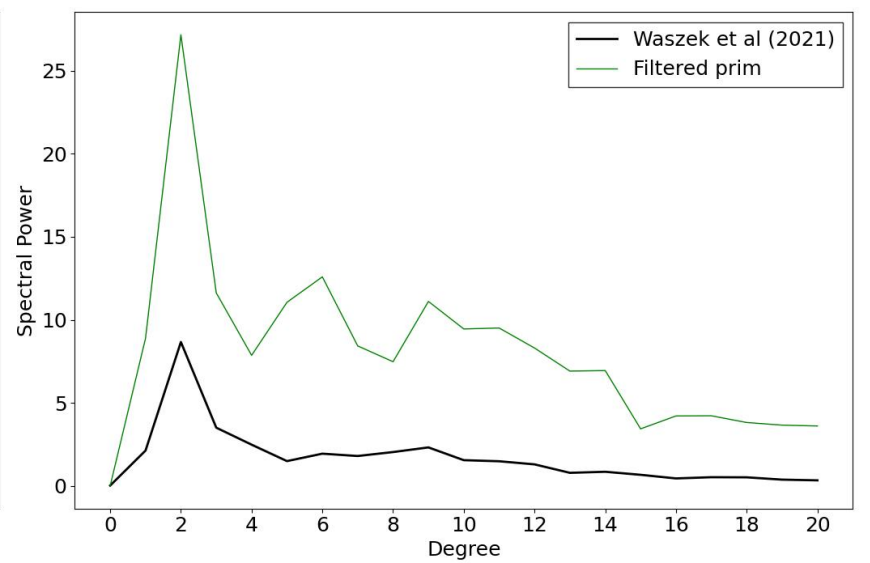
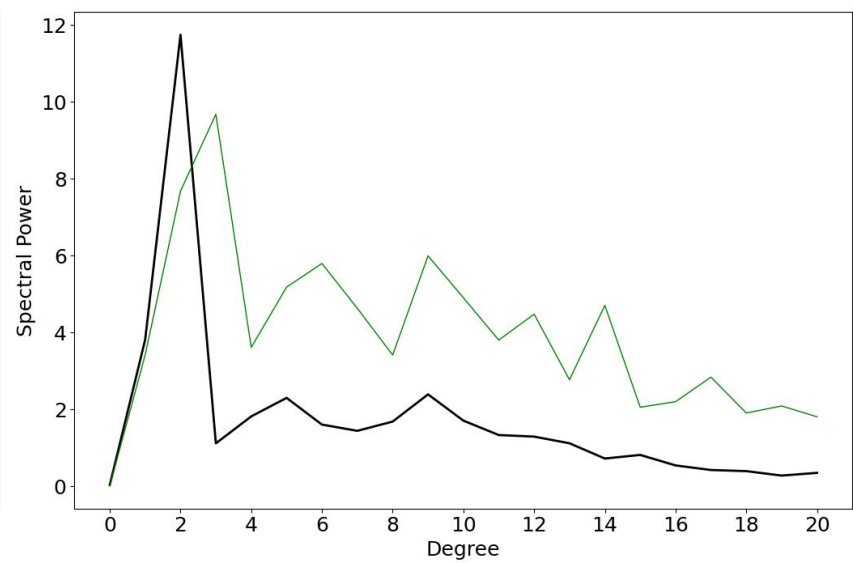
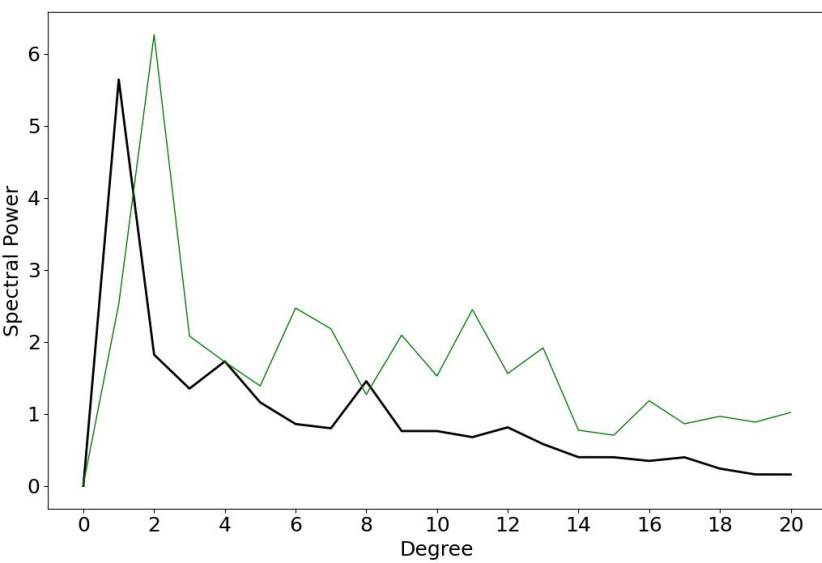
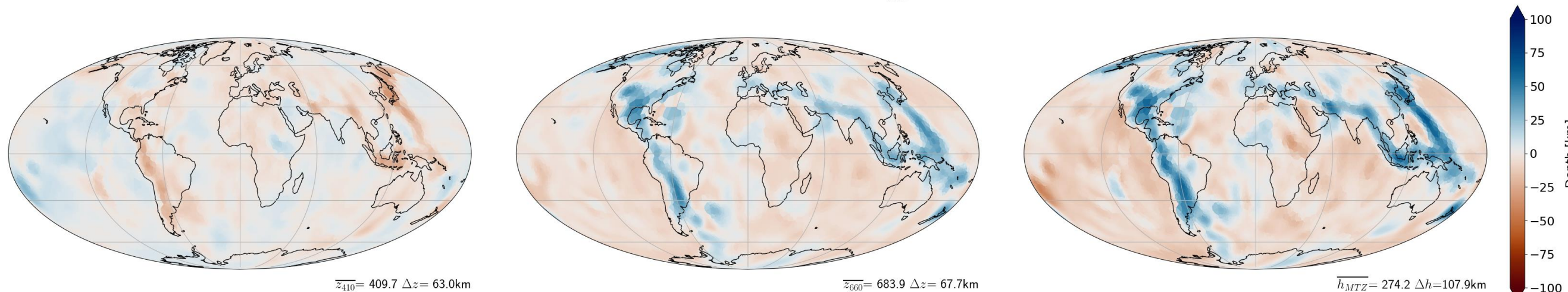
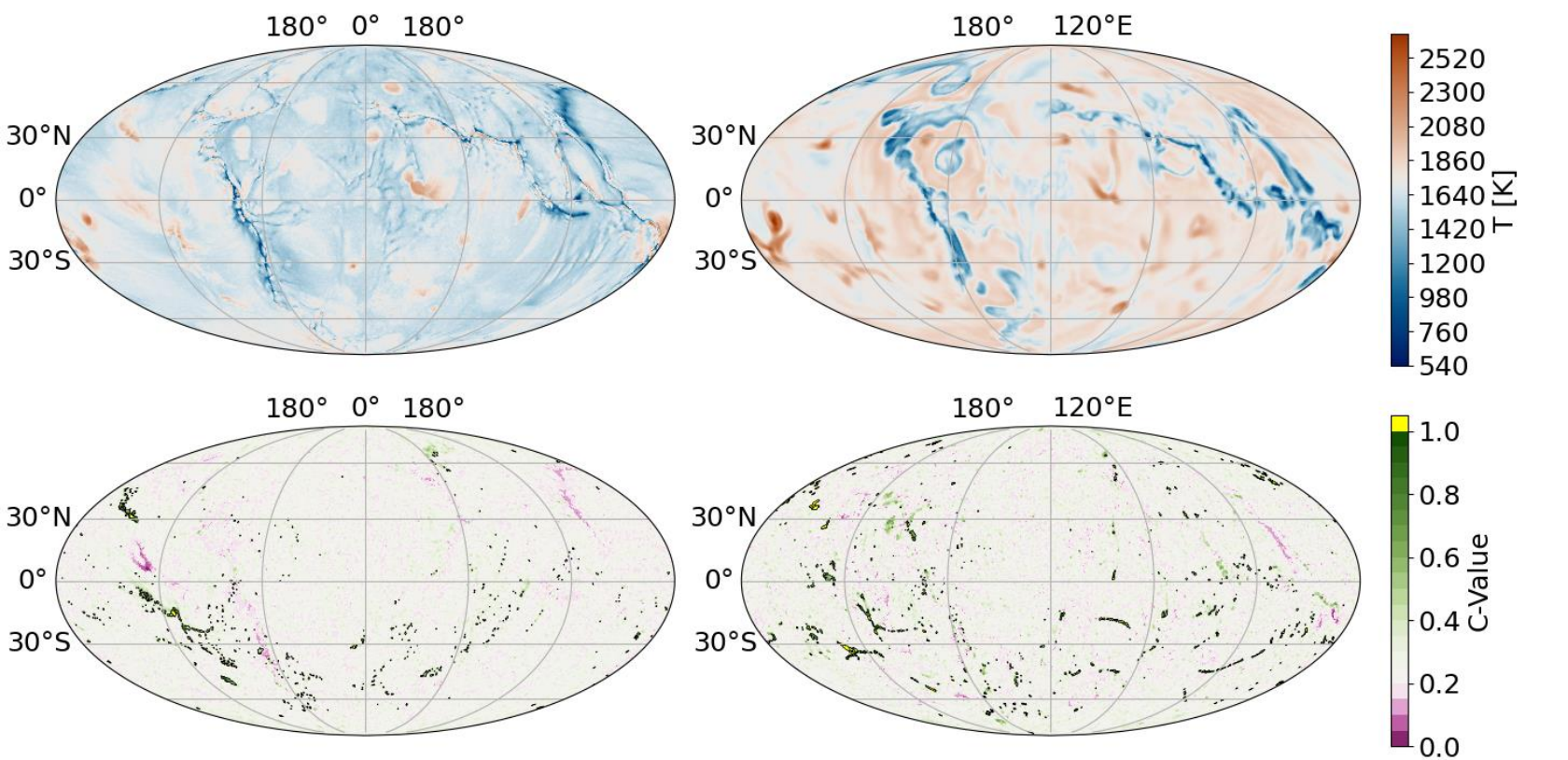
Waszek et al (2021) Bouncepoints



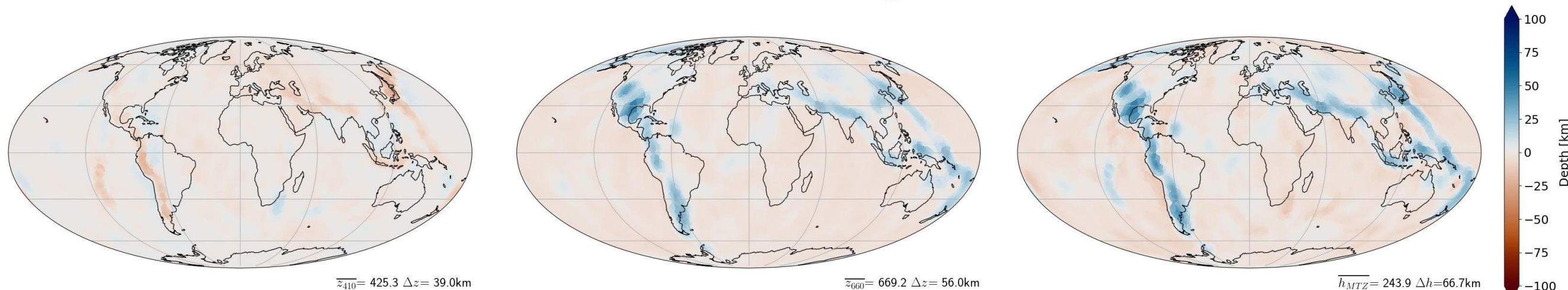
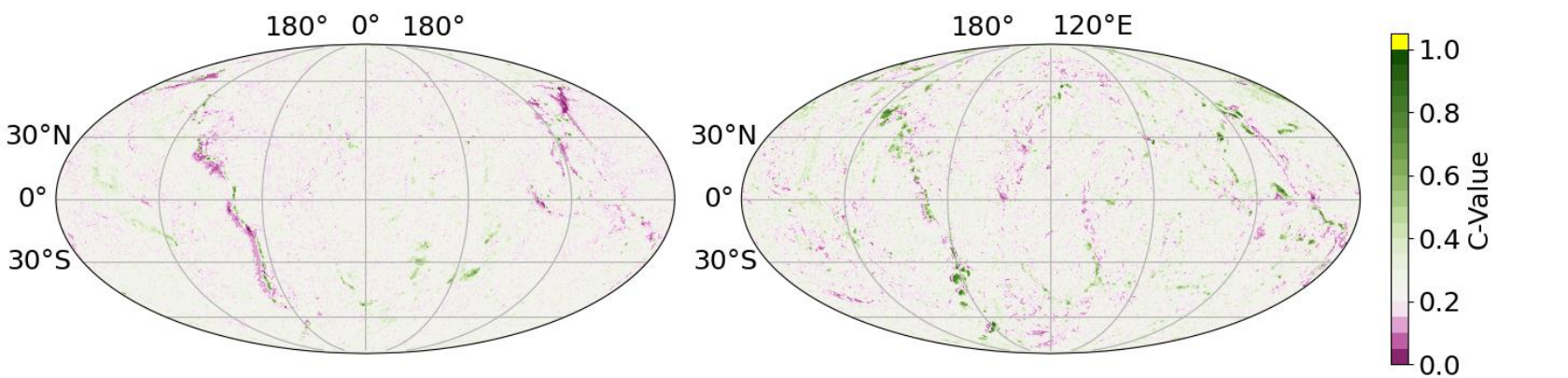
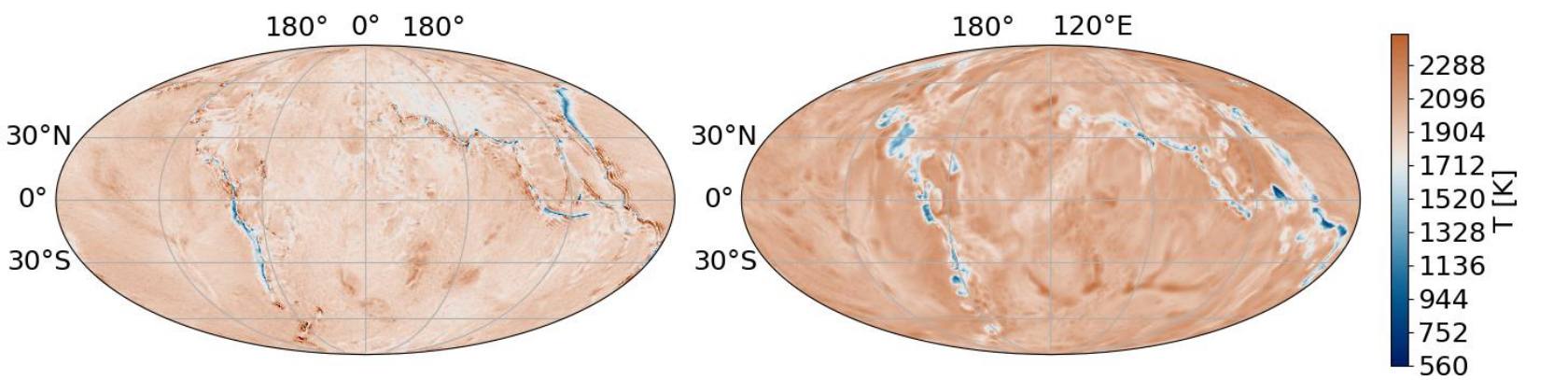
Reference Simulation



Primordial Simulation



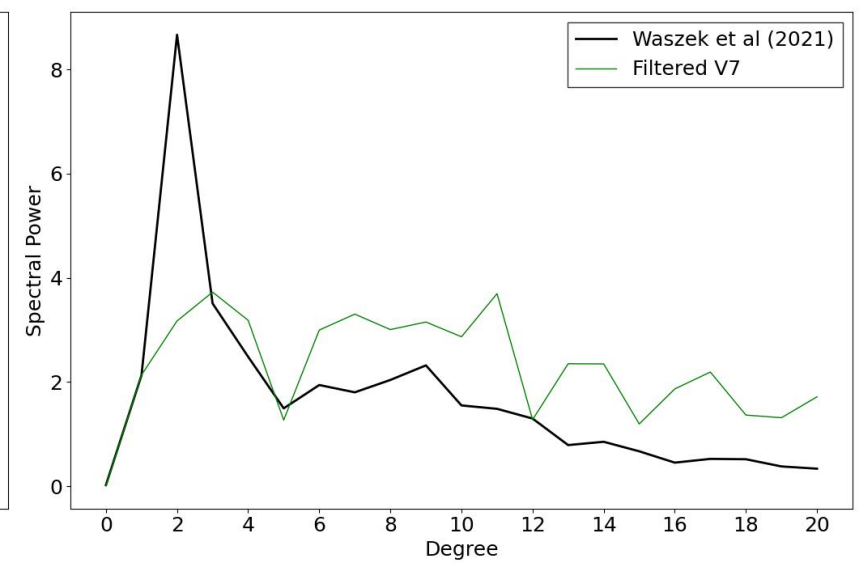
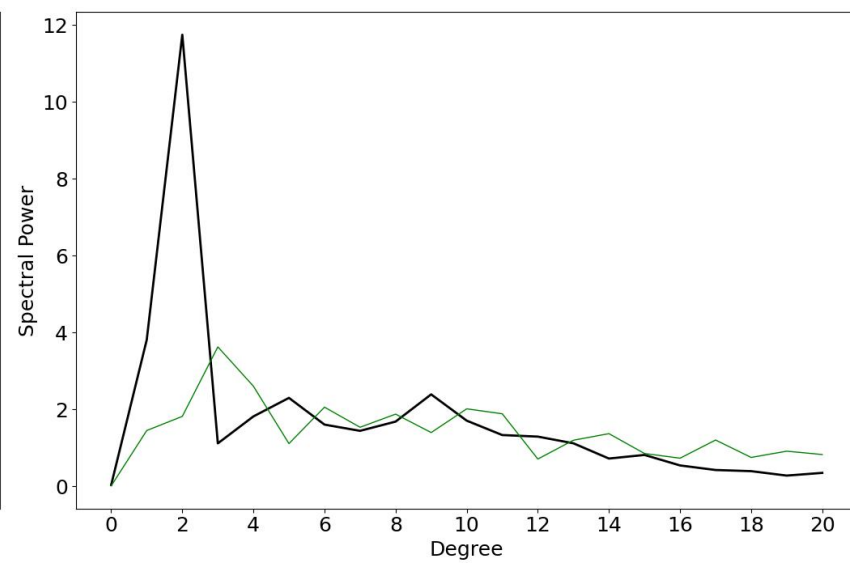
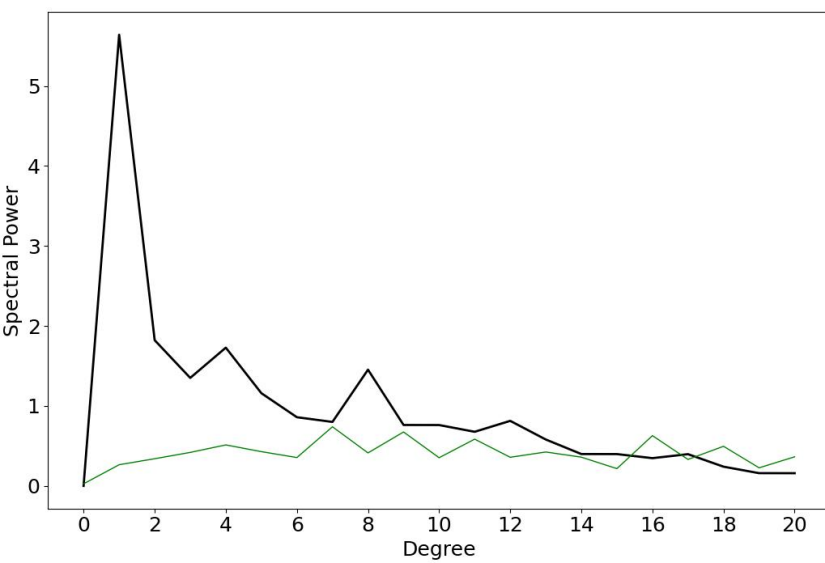
V7 Simulation



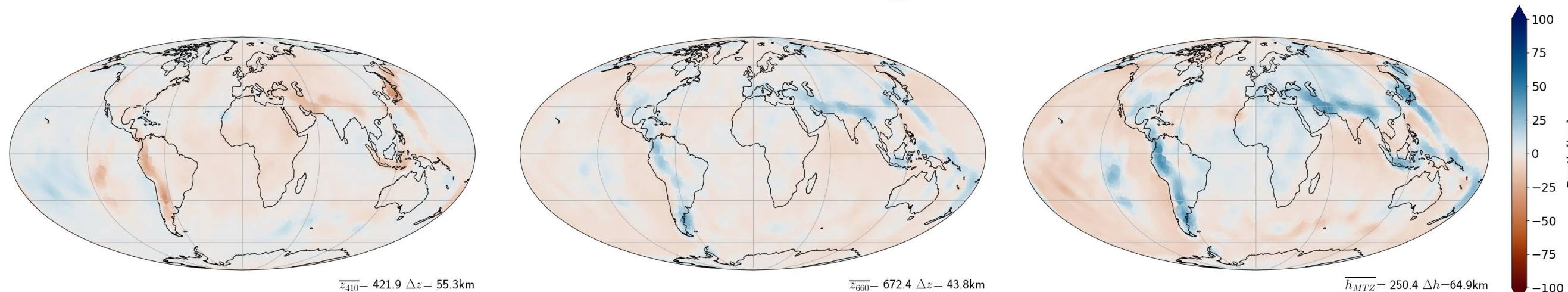
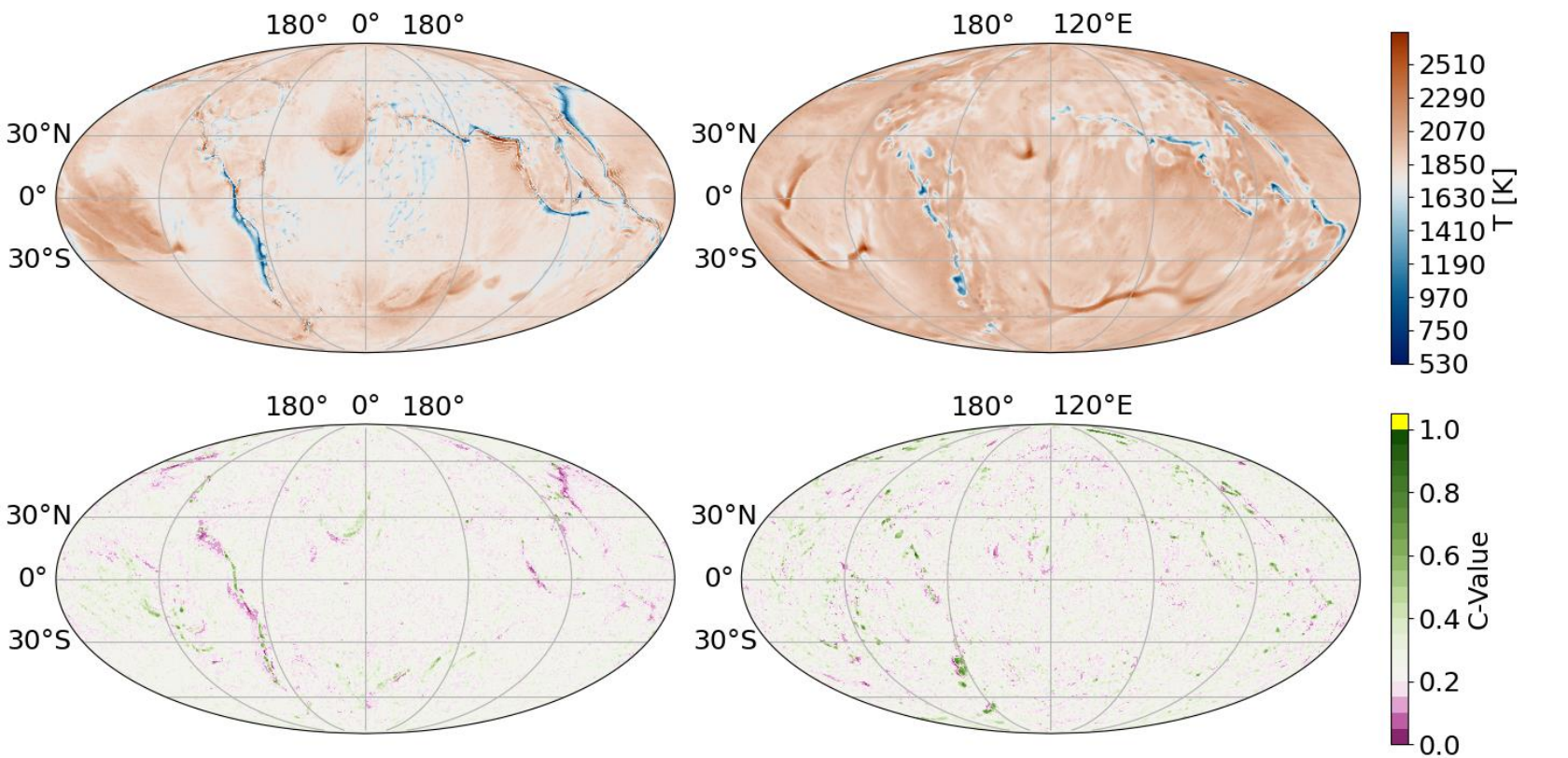
$\overline{z_{410}} = 425.3 \Delta z = 39.0 \text{ km}$

$\overline{z_{660}} = 669.2 \Delta z = 56.0 \text{ km}$

$\overline{h_{MTZ}} = 243.9 \Delta h = 66.7 \text{ km}$



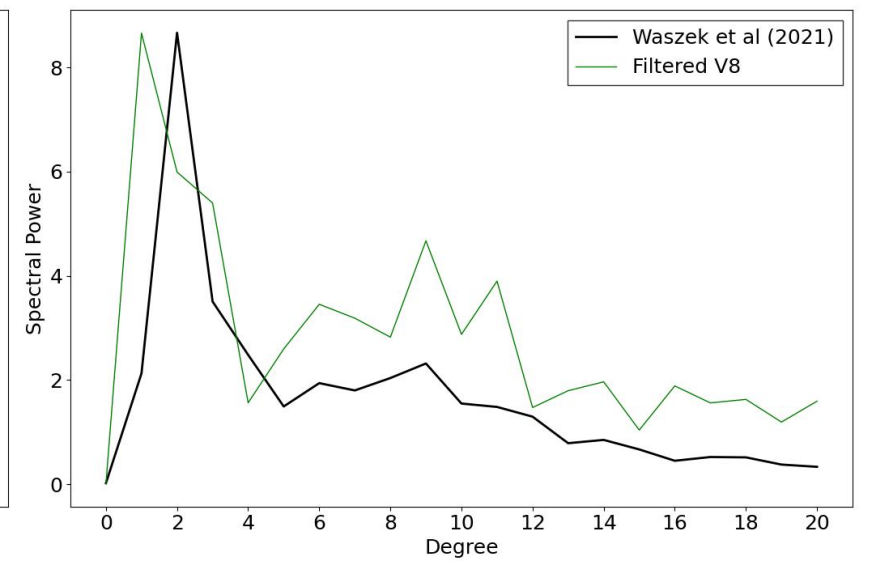
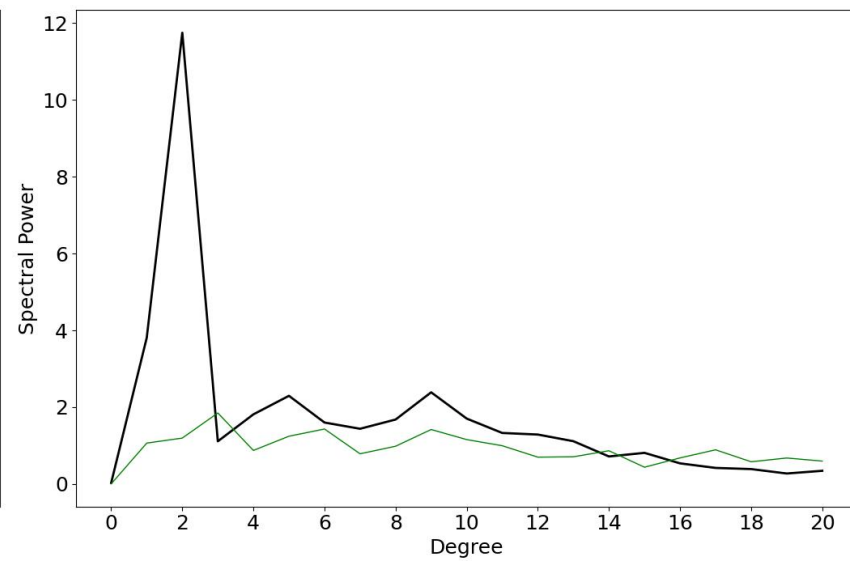
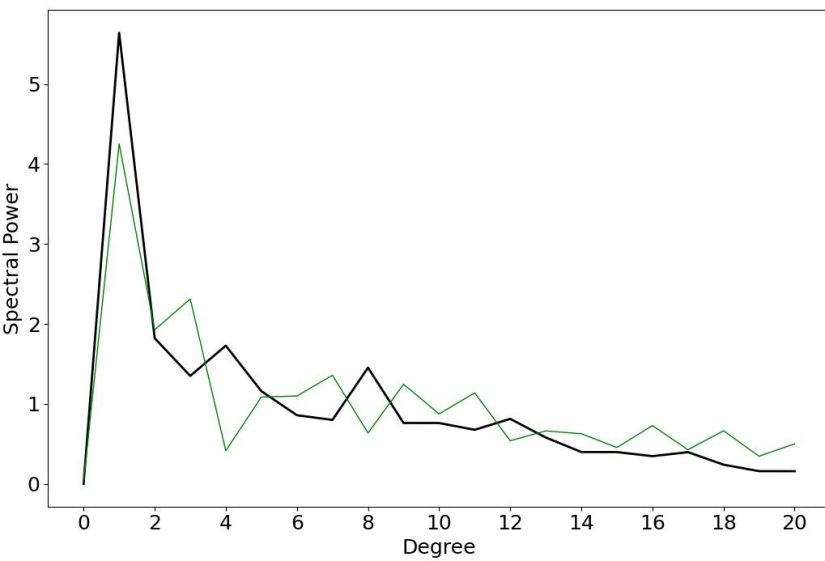
V8 Simulation



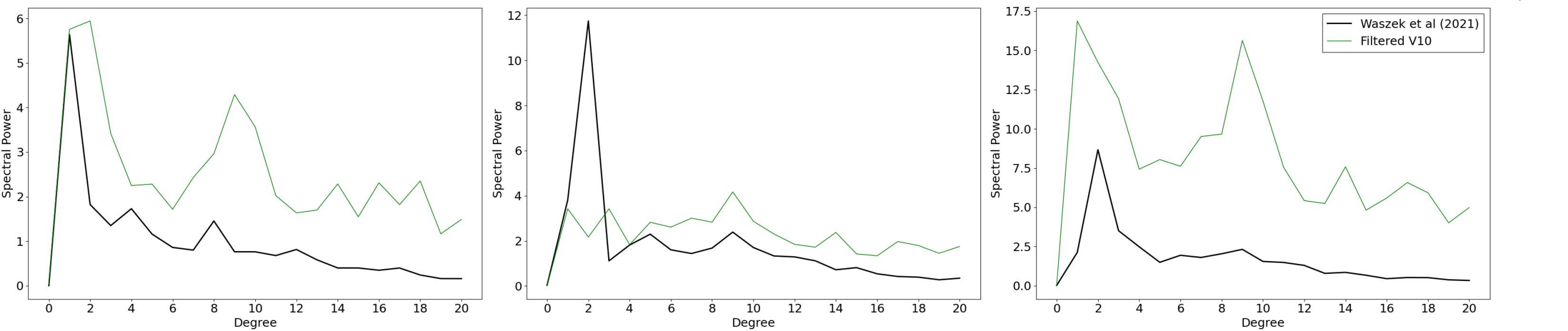
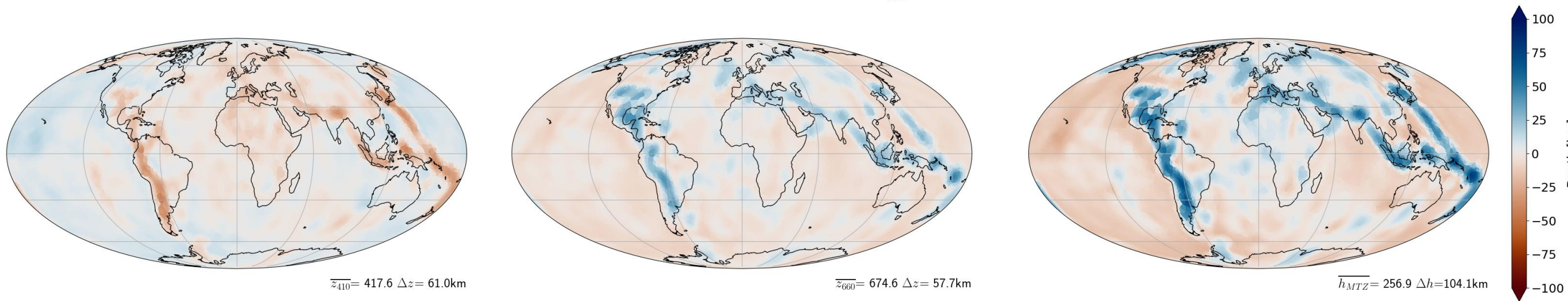
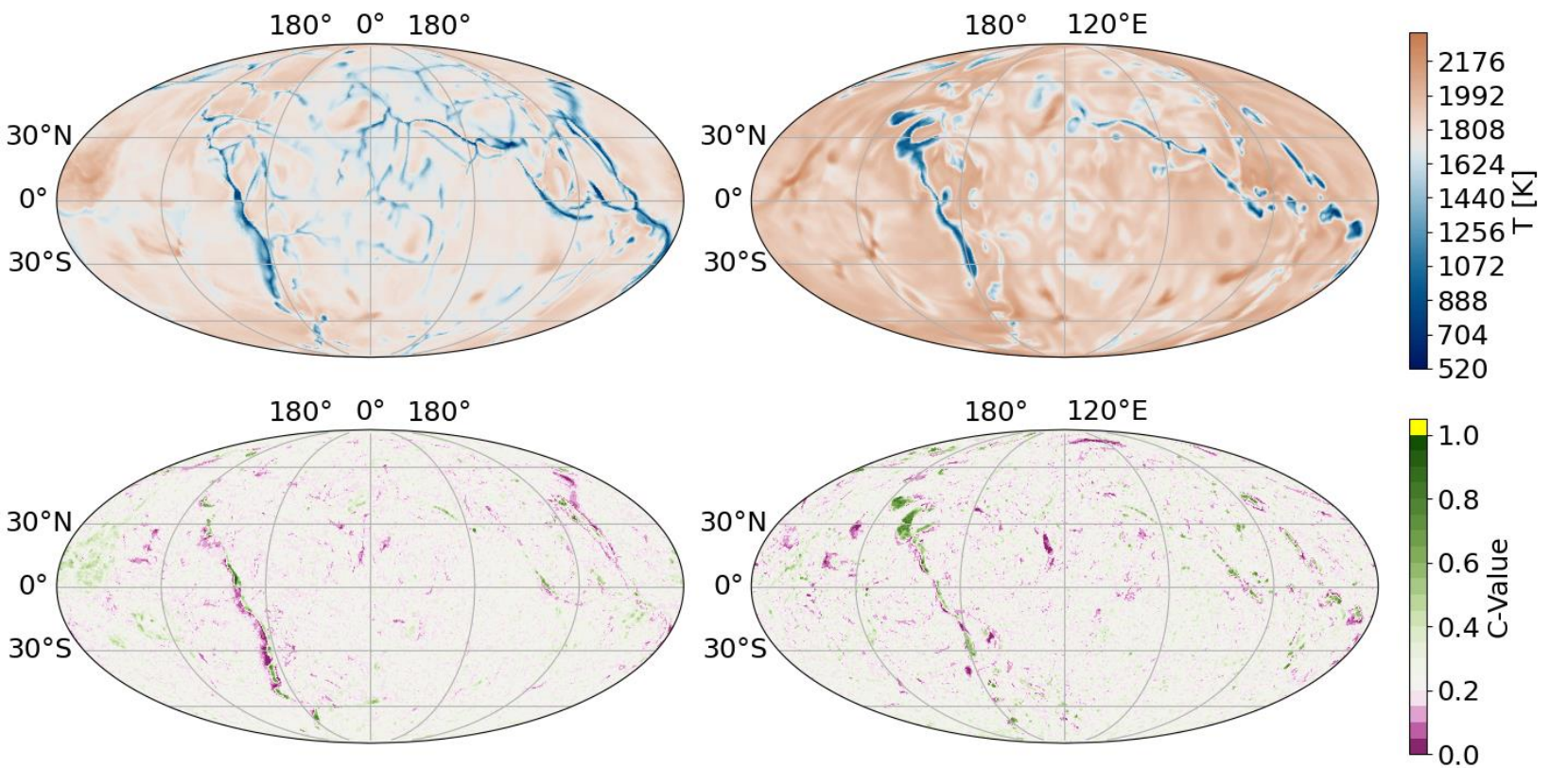
$\overline{z_{410}} = 421.9 \Delta z = 55.3 \text{ km}$

$\overline{z_{660}} = 672.4 \Delta z = 43.8 \text{ km}$

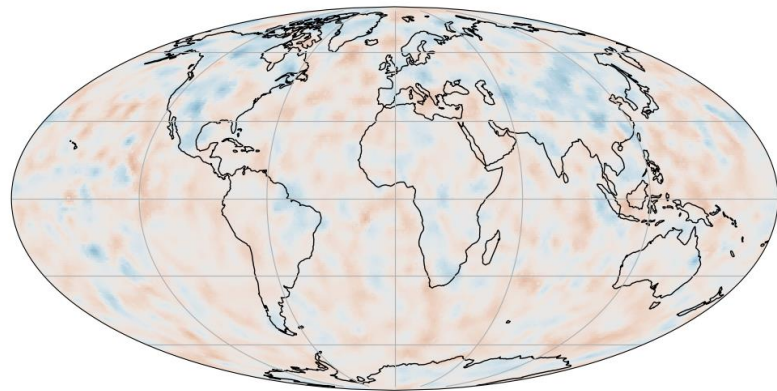
$\overline{h_{MTZ}} = 250.4 \Delta h = 64.9 \text{ km}$



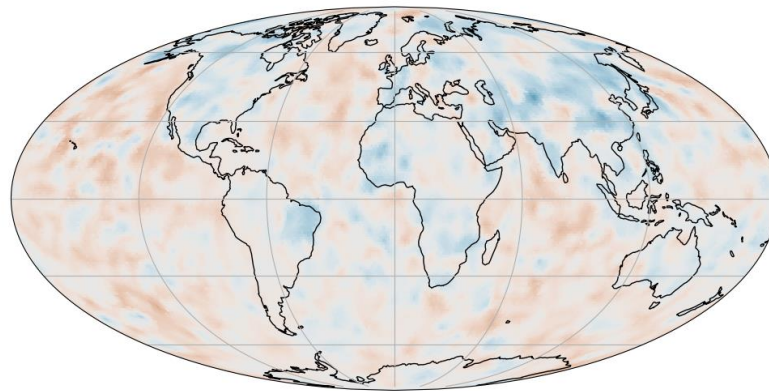
V10 Simulation



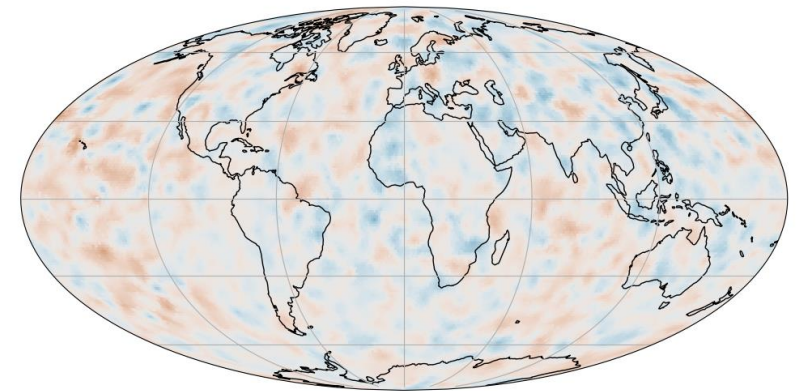
What if we compare to other MTZ topos?



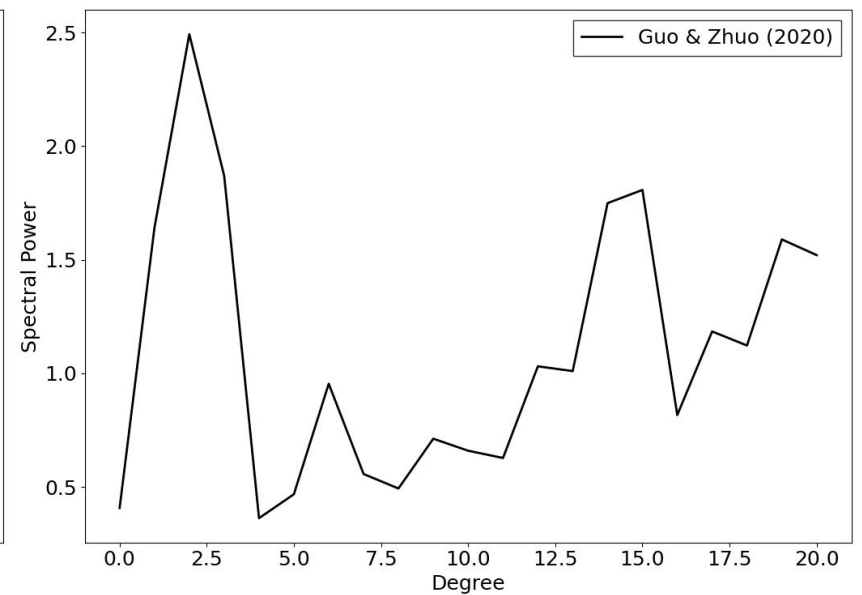
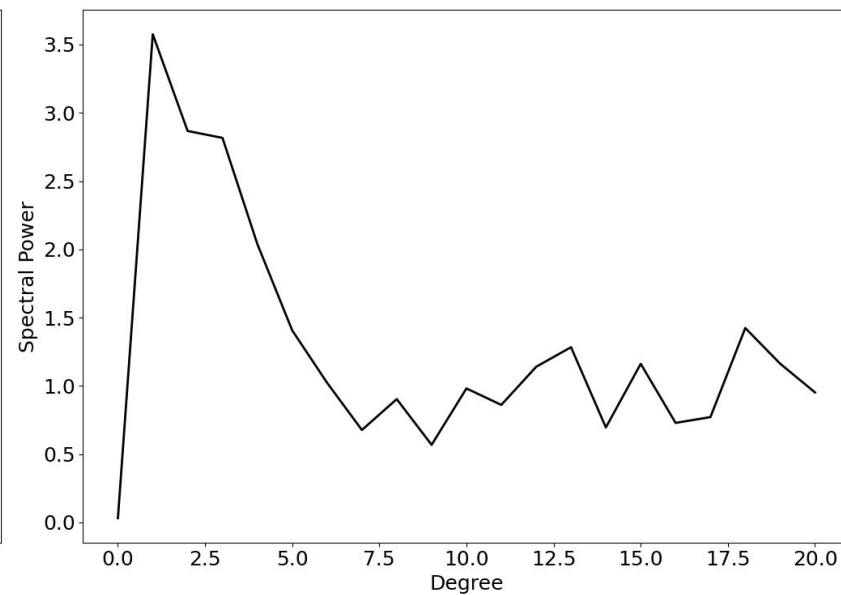
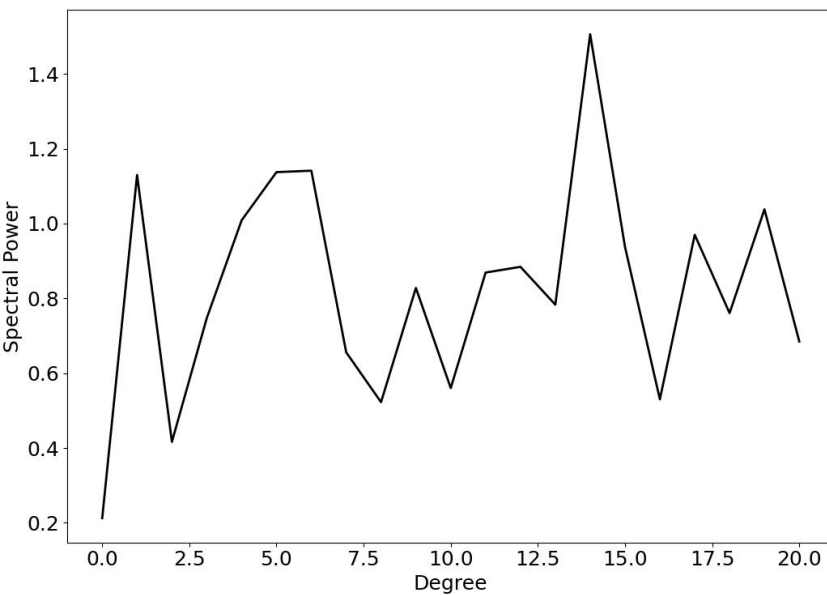
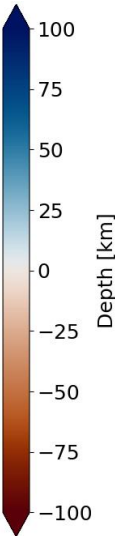
$\overline{z_{410}} = 411.6 \Delta z = 37.5 \text{ km}$



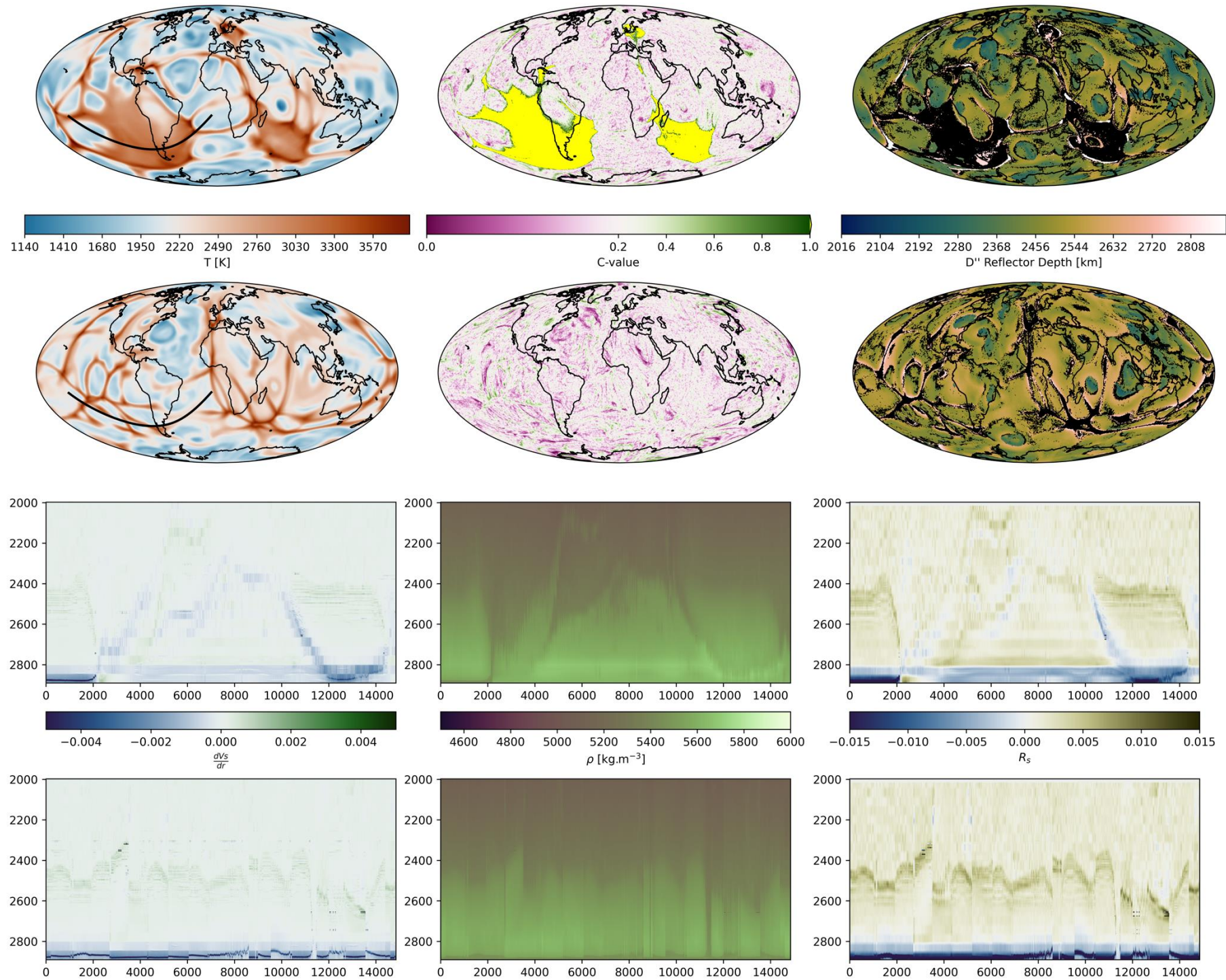
$\overline{z_{660}} = 660.4 \Delta z = 43.3 \text{ km}$



$\overline{h_{MTZ}} = 248.8 \Delta h = 44.5 \text{ km}$



D'' using ceb tables



D'' using fsp tables

